



1. Core (Mandatory) ML Tools

These are **non-negotiable** for Python ML.

Python

- Version: **Python 3.9+**
- Why: Main ML ecosystem

Package / Environment Management

Choose one:

- venv (simple, built-in)
- Conda (popular in data science)

Reason: To ensure dependency isolation, which is very beneficial for IT purposes.

2. Core Python ML Libraries

This is the **foundation stack**.

NumPy

- Numerical computation
- Arrays, matrices, statistics

Pandas

- Data loading and cleaning
- CSV, JSON, Excel

Matplotlib (or Seaborn)

- Data visualisation
- Debugging data issues

Scikit-learn

- Classical ML algorithms
- Preprocessing
- Model evaluation

If you know these four, you can do **real ML**.



3. Development and Productivity Tools

These improve workflow (essential for teaching).

Jupyter Notebook / JupyterLab

- Interactive learning
- Visualisation
- Experiments

IDE

Pick one:

- VS Code (most common)
- PyCharm

4. Data Handling and Storage

Common in Real Projects:

- CSV / JSON files
- SQL databases (PostgreSQL, MySQL)
- Object storage (S3-compatible)

Focus on teaching how to **read and write data**, rather than on database theory.

5. Model Persistence and Experiment Tracking

Introduce after the basics.

Joblib / Pickle

- Save and load models

MLflow (optional but powerful)

- Track experiments
- Model versions
- Parameters and metrics

6. Deep Learning Tools

Only after classical ML.

PyTorch or TensorFlow

Pick **one**:

- PyTorch is more popular for learning
- TensorFlow is common in production

GPU support (optional)

- CUDA (if available)
- Cloud GPUs (later)

7. MLOps and Deployment Tools

Introduce at the end.

Docker

- Package ML apps
- Reproducibility

Kubernetes (advanced)

- Scaling ML services

API Frameworks

- FastAPI
- Flask

Serve models as REST APIs.

8. Version Control and Collaboration

Absolutely required.

Git

- Code versioning
- Collaboration

GitHub / GitLab

- Repositories
- CI/CD (later)



9. Monitoring and Maintenance

For Production ML Systems:

- Prometheus / Grafana (metrics)
- Data drift monitoring
- Logging (ELK stack)

10. Minimal Setup Checklist

If you want a **simple starting setup**, this is enough:

Python 3.9+

venv or conda

NumPy

Pandas

Scikit-learn

Matplotlib

Jupyter Notebook

Git

Tools change. **Concepts don't.**